

LOT (R) COMPUTER

ROADMAP + RISK REGISTER -- 3 July 2026

Document: LOT-COMPUTER-ROADMAP.md

Reads together with: VISION, RIG-SPEC, FIRMWARE, SOFTWARE-BRIDGE (all in this folder / docs/corporate)

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00 – Phased Plan

- PHASE 0 – Paper (this session) Complete 2026-07-03
- Vision document DONE
 - Hardware BOM + supplier links DONE
 - Firmware specification DONE
 - Software / API connector specification DONE
 - This roadmap DONE
- PHASE 1 – Fit-Check Prototype (breadboard + 3D-printed shell mockup)
- Confirm ESP32-S3 + BME680 + piezo disc bring-up on a dev board
 - 3D-print a 40x40x5mm shell mockup to physically test the
 - camera-boss decision (RIG-SPEC §03) before cutting real steel
 - Battery fit-check: does 150mAh fit, or does the tile need to drop
 - to a 40-80mAh bridge cell (RIG-SPEC §07)
 - Order 1x PCBWay CNC stainless sample shell (not the full run) to
 - verify tolerances against the fit-check
 - GATE: no full order placed until fit-check resolves §03 and §07
- PHASE 2 – Backend Enablement
- Build the Device model + bearer-token middleware
 - (SOFTWARE-BRIDGE §03) – single contained PR
 - Stand up GET /notifications/next + wire POST /logs event type
 - "device.copy" / "device.session"
 - GATE: device-token auth code-reviewed and merged before any unit ships with live credentials
- PHASE 3 – First Article (1-5 units)
- Full PCBWay order: PCB + SMT assembly + CNC stainless shell,
 - qty 5, not 100 – validate the real supply chain end to end
 - Firmware bring-up against Phase 2's live backend, on real hardware
 - OTA rollback test (FIRMWARE §06) – deliberately push a broken image,
 - confirm A/B rollback recovers without a bench connection
 - GATE: 5 units running for 2+ weeks on a charging puck with zero unintended camera activation, zero cross-session data leakage
- PHASE 4 – 100-Unit Pilot Run
- Place full BOM order (RIG-SPEC §10) – ~\$8,525 unit cost + \$2-4k NRE
 - PDF manuals finalized and included per unit (see §02 below)
 - GATE: first 10 units held back as a burn-in batch before the remaining 90 ship
- PHASE 5 – Field Data Review
- 30/60/90-day review of real Copy-tap rates, weather-sensor drift, and OTA success rate feeds back into whether v2 revisits the camera-boss decision (§03) or the PM2.5 dock decision (RIG-SPEC §04)

01 – Risk Register

#	Risk	Where flagged	Mitigation / Decision Path
1	OV2640 lens assembly does not fit flush in 5mm body	RIG-SPEC §03	Accept lens boss (Option A) for v1; re-evaluate bare-
2	150mAh cell alone may consume most of the height budget	RIG-SPEC §07	Reframe as bridge cell for a dock-resident device,
3	Good Display e-paper panel is quote-only, not retail-priced	RIG-SPEC §02	Get formal 100-unit quote before Phase 4 order
4	No device-auth backend exists today	SOFTWARE-BRIDGE §03	Phase 2 is a hard gate – no device ships with live credentials
5	Camera privacy – any camera on a device sitting in someone's home is a trust risk regardless of intent	FIRMWARE §02, §03	
6	Single-vendor manufacturing (PCBWay for CNC + PCB + assembly)	RIG-SPEC §09	Acceptable for a 100-unit pilot where audita
7	OTA bricking in the field	FIRMWARE §06	A/B partition rollback tested explicitly in Phase 3 before Phase 4 volume order
8	E-paper ghosting / display legibility degrading over many partial refreshes	FIRMWARE §04	Automatic full-refresh cycle e

Risk 5 is the one worth stating plainly: a camera in a home product is a

trust liability first and an engineering problem second. The firmware invariant (never serialize a frame) is necessary but not sufficient – the PDF user manual (§02) must say this in plain language on page one, not bury it in a firmware doc nobody but engineering reads.

02 – Documentation & PDF Manuals

Per the intake brief's requirement that firmware, software, and hardware stay **separate documents**, and that the result includes **PDF manuals**:

```
| Manual | Source | Audience |
|---|---|---|
| Quick Start / User Manual | This vision doc + rig-spec highlights, condensed | End user, printed insert in the box |
| Firmware Manual | `LOT-COMPUTER-FIRMWARE.md`, as-is | Engineering / OTA release process |
| Software / API Manual | `LOT-COMPUTER-SOFTWARE-BRIDGE.md`, as-is | Backend engineering, companion-app developer |
```

PDF generation follows the existing repo pattern used for the Badges & Achievements Codex (**`scripts/generate_badge_pdf_v19.py`**, reportlab-based, Terminal Grid typography) rather than introducing a new toolchain. See **`docs/technical/pdf/`** for the generated set produced alongside this roadmap.

03 – 100-Unit Run – Why 100, Not 10 or 1,000

- **10 units** would not produce enough field-hours to validate the camera-boss decision, the bridge-cell battery sizing, or OTA rollback under real (not bench) conditions.

- **1,000 units** commits to injection-molding-scale tooling and a BOM that has not yet cleared a single fit-check prototype (Phase 1) – an unjustifiable order of operations.

- **100 units** matches the precedent already set for LOT® Station and LOT® Brush hardware cohorts ([docs/corporate/LOT-AMBIENT-AI-VISION.md](#)) – enough to seed a real Usership hardware cohort, small enough that a Phase 1/3 mechanical surprise doesn't strand five figures of unusable stainless steel.

04 – Cost Rollup (from RIG-SPEC §10)

100-unit BOM subtotal	≈ \$8,525
NRE (tooling, fixtures, article)	≈ \$2,000 - \$4,000
Pilot run total	≈ \$10,500 - \$12,500
Per-unit landed cost	≈ \$105 - \$125

No retail price is set in this document – that is a commercial decision outside engineering scope. For reference only, the COSMO® robotics roadmap ([docs/corporate/LOT_ROBOTICS_COSMO.md](#)) prices the Quantum Cube-class hardware precedent in the \$2,500-\$5,000 range for a different product tier; the LOT® Computer, as a simpler pager-class device, is not assumed to share that price point.

05 – Immediate Next Action

Phase 1 (fit-check prototype) is the only phase that should start without further sign-off – it is cheap (a dev board, a 3D print, one CNC sample shell) and resolves the two open engineering risks (§01, items 1 and 2)

before any money is committed to a 100-unit order. Phases 2-4 each carry an explicit gate above and should not be started out of sequence.

LOT® Founded 7 April 2016 · COSMO® Founded 1 July 2024

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